

MAHDI ASEFI

+98 910 011 1974 asefi.mahdii@gmail.com
Alborz, Iran mahdiAsefi20
mahdiasefi

EDUCATION

- 2023 – Present **M.Sc. in Artificial Intelligence** Azad University | Tehran, Iran
GPA: 4 of 4
Supervisors: Prof. Azadeh Mansouri, Prof. Reyhaneh Tati
- 2018 –2023 **B.Sc. in Computer Science** University of Kashan | Isfahan, Iran
GPA: 3.4 of 4
Supervisor: Prof. Mahsa Soheil Shamaee
• Received a stipend for exceptional performance in the undergraduate project titled: "Persian Paraphrasing"

RESEARCH INTEREST

- **General:** Artificial Intelligence, Computer Vision, Machine Learning, Deep Learning
- **Specific:** Deepfake face detection, Deepfake Generation, GANs in Medical Imaging

PROGRAMMING LANGUEGES AND TECHNOLOGIES

- Python • C# • TensorFlow • PyTorch • C++ • Unity • Java
- scikit-learn • Matplotlib • Git • Huggingface Transformers • OpenCV

RESEARCH ASSISTANT

- 2024 – Present **Deep Fake Detection** Supervisor: Prof. Azadeh Mansouri
- Reviewed and synthesized insights from five deepfake detection papers.
 - Presented findings and methodologies to professors and team members, facilitating discussions.
 - Conducted experiments by training models proposed in two of the reviewed papers.
 - Evaluated model performance and limitations through rigorous testing.
 - Developed an innovative approach to combine methods from the papers for improved face deepfake detection.
- Winter 2023 **GANs in Radiogenomics** Supervisor: Prof. Qian Liu
- Studied the use of GANs to generate synthetic data for breast cancer detection, combining MRI images and genomics information.
 - Investigated techniques to fuse genomics data with MRI feature maps, creating a unified network to generate new images.
 - Analyzed and presented multiple research papers on GAN-based data augmentation for improving radiogenomics models.
 - Focused on enhancing data availability for machine learning models through synthetic data generation from real medical imaging and genomic inputs.
- March 2022 **Authentication and Clustering People in News** Supervisor: Dr. Pooria Lajevardy
- Summarized and translated key points from two papers on face detection and identification in news images.
 - Analyzed limitations in the methodologies and provided feedback and suggestions.
 - Discussed potential solutions and alternative approaches with the team.

TEACHING ASSISTANT

Spring 2019

Fundamentals of Computer and Programming

Professor: Prof. Mahsa Soheil Shamaee

- Designed assignments to enhance students' understanding of programming basics in C++.
- Taught C++ programming concepts and provided hands-on coding experience using Visual Studio.
- Assisted approximately 10 students with problem-solving and debugging during coding sessions.
- Graded assignments and provided feedback to help students improve their programming skills.

WORK EXPERIENCE

Jan 2026
– Present

AI Engineer

Metadeep | Karaj, Alborz

- Architected a multi-model AI pipeline integrating several video generation APIs with a model-agnostic orchestration layer.
- Built a Vision LLM asset analysis engine to extract visual properties and auto-generate structured creative briefs.
- Engineered a prompt compilation system translating creative briefs into shot-level generation prompts across multiple cinematic styles.
- Integrated a Persian/Farsi voice-over pipeline using voice cloning and TTS with RTL subtitle rendering.
- Conducted systematic R&D benchmarking of open-source and commercial generative video models.
- Performed frame-level video analysis to extract reusable cinematic style guides and prompt templates.

ComfyUI / FFmpeg / Vision LLM / ElevenLabs / FastAPI / LangChain / Redis

Sep 2025
– Jan 2026

AI Engineer

Tetronix | Remote (Canada)

- Developed a scalable web crawling pipeline for aggregating car marketplace data with MongoDB.
- Implemented similarity-based vehicle retrieval system for related-car search.
- Designed and deployed FastAPI backend services for data processing and AI APIs.
- Built an AI-powered chatbot using OpenAI API with structured prompts and Langfuse observability.
- Implemented short-term (Redis) and long-term memory via message summarization.
- Integrated real-time voice interaction using ElevenLabs.

MongoDB / FastAPI / OpenAI API / Langfuse / ElevenLabs / Redis

2023
– 2025

AI Engineer

Drivee.ai | (remote), US

- Joined early and contributed to computer vision and LLM products: Drivee AI – Drivee Sales Agent – HeyGent.
- Built high-accuracy foreground segmentation and car part detection models (YOLO).
- Developed lighting enhancement, depth-based bokeh, and CGAN-based shadow/reflection systems.
- Created AI chatbots and multi-agent systems for sales, booking, and customer engagement.
- Integrated LLM solutions with CRM, DMS, and scheduling tools.

Python / PyTorch / OpenAI API / YOLO / GANs / OpenCV / Multi-Agent Systems / LLM

Fall 2022

NLP Engineer Intern

Commercialization and Innovation Center of University of Kashan | Isfahan, Iran

- Implemented NLP models for Question Answering using Hugging Face.

PyTorch Lightning / Transformers

- Achieved 99% accuracy in long-distance car detection from UAV aerial images.
- Built and deployed real-time car detection & tracking system on server.

Python / PyTorch / YOLO

PROJECTS

Undergraduate
Project

Persian Paraphrasing

[Github](#)

- Developed a Persian paraphrasing model using state-of-the-art transformer models.
- Translated English datasets to Persian due to the lack of existing Persian paraphrasing data.
- Overcame translation challenges by cleaning and curating data.
- Fine-tuned a multilingual T5 model, achieving strong paraphrasing and error correction.
- Received a stipend for exceptional research contributions.

AI Course
Final Project

Genetic and PSO algorithms

[Github](#)

- Implemented Genetic and Particle Swarm Optimization for an optimization problem using Python.

LANGUAGES

English: [TOEFL Score 91](#) • Reading: 24 • Listening: 24 • Speaking: 20 • Writing: 23

Persian: Native

REFERENCES

• [Prof. Azadeh Mansouri](#)

• [Prof. Mahsa Soheil Shamaee](#)

• [Prof. Saeed Aaseedi](#)

• Email: a_mansouri@khu.ac.ir

• Email: soheilshamaee@kashanu.ac.ir

• Email: aseedi@kashanu.ac.ir